

Visual Prompting for Region-Grounded Captioning of Pulmonary Nodules in CT Images Using VLMs

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Abstract. Deep learning-based methods have been widely studied for benign-malignant discrimination of pulmonary nodules on CT, but many lack explainability because they provide only final classification results. Vision-language models can generate descriptive imaging findings; however, in full-slice CT images, attention may shift from the target nodule to nodule-like structures such as vessels or bronchi. This study investigates rectangular visual prompting for region-grounded pulmonary nodule captioning. Using LIDC-IDRI structured morphology annotations, we constructed an image-text dataset with standardized findings and trained a captioning model based on a vision transformer and open-source large language models, including GPT-oss. Evaluation using captioning metrics, morphology-level agreement, and attention visualization showed that rectangular prompts improved finding generation and spatial focus.

Keywords: Image-to-Text, Pulmonary Nodules, Visual Prompting.

1 Introduction

Lung cancer is a major cause of cancer-related death, and low-dose CT screening has increased pulmonary nodule detection. While deep learning-based classification methods are widely studied, many lack explainability because they output only final predictions. Vision-language models can generate descriptive CT findings but may attend to nodule-like structures rather than the target nodule in full-slice images. This study evaluates rectangular visual prompting for region-grounded pulmonary nodule finding generation compared with unprompted full-slice CT images.

2 Methods

In this study, a proprietary dataset was constructed based on nodule region extraction and structured data derived from chest CT images within the LIDC-IDRI dataset. Images were input to the vision model, and the extracted features were projected into open-

source large language models, including GPT-oss [1], to generate findings. Performance was evaluated using both automatic text-generation metrics and clinically oriented assessments. Specifically, we computed standard captioning scores against reference findings and quantified agreement of key morphological attributes between the generated findings and the structured annotations. Additionally, we visualized the model's attention to determine if the generated findings were spatially grounded in the nodule region.

3 Results

The visual prompt improved finding generation compared with unprompted full-slice CT images. In the ViT + GPT-oss 20B model, BLEU-4 increased from 0.631 to 0.674, and CIDEr increased from 3.322 to 3.833. Morphology-level evaluation showed improved agreement across all evaluated features, including sphericity, margin, and texture. Attention visualization demonstrated that the prompt guided the model's focus toward the target nodule and reduced attention dispersion to surrounding structures as shown in Fig.1.

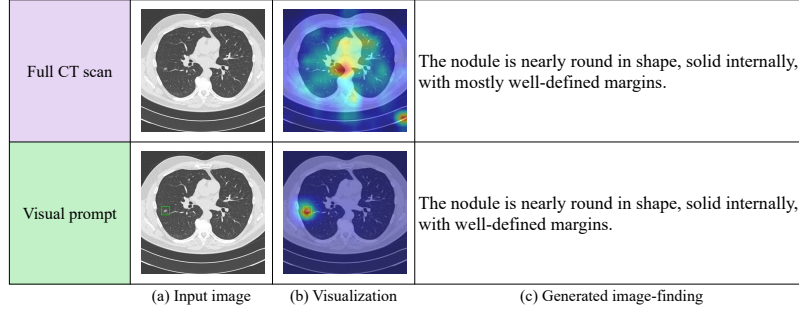


Fig. 1. Examples of the generated image findings and attention visualizations.

4 Conclusion

In this study, we proposed a rectangular visual prompting approach for region-grounded findings generation from chest CT images. These results suggest that a simple visual prompt can provide a practical way to improve the reliability and interpretability of CT findings generation.

References

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